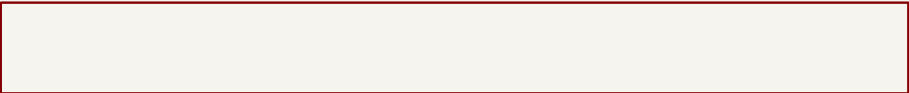


ALFRED / A1-4RD

5 V USB-C PD | removable Pi Zero 2 W | full-duplex I2S | one jaw servo

USB-C: fixed 5 V / 5 A contract



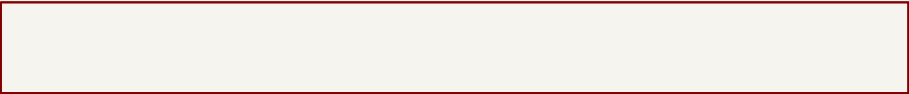
File: block1.kicad_sch

Protected rails and servo undervoltage override



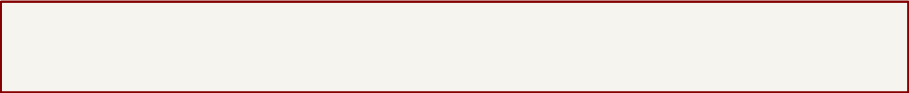
File: block2.kicad_sch

Raspberry Pi physical header and shared I2S



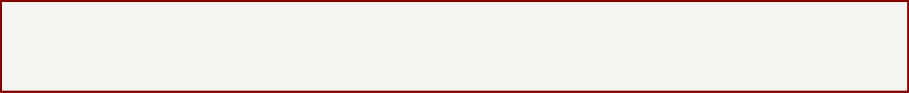
File: block3.kicad_sch

Digital microphone and bridge speaker amplifier



File: block4.kicad_sch

Jaw interface, energy dump, button and status



File: block5.kicad_sch

Prototype documentation: Design-notes.md, BOM.csv, Mechanical-interface.md

Routed Alfred carrier. Unbuilt prototype; follow revision C bring-up checks.
A1-4RD / Alfred

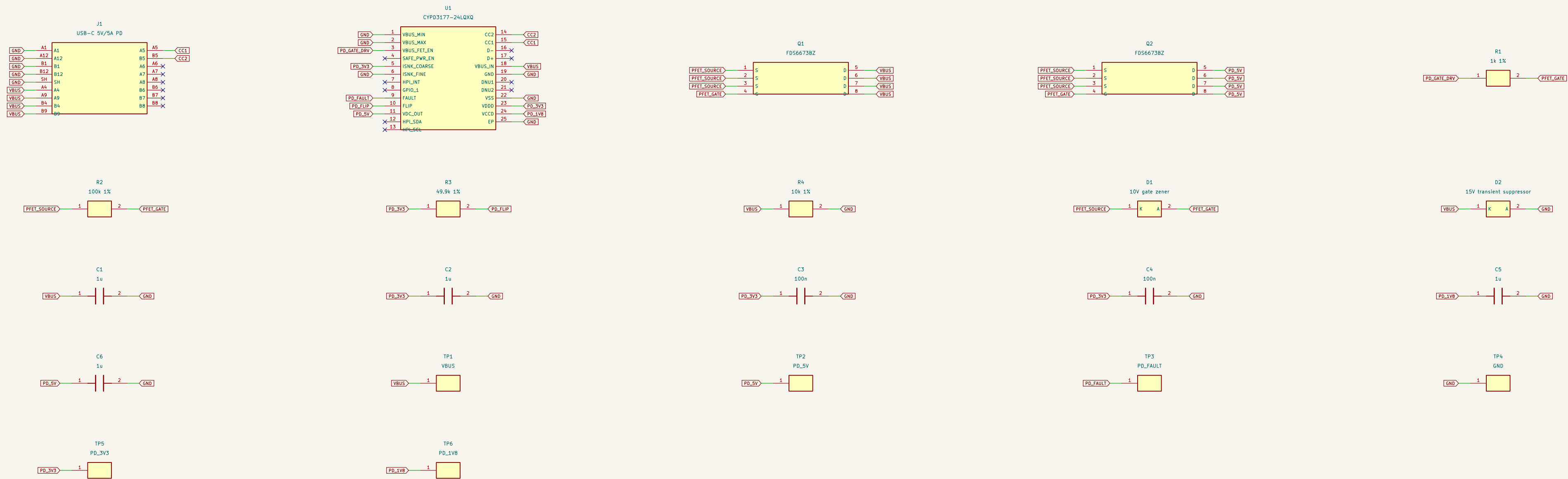
Sheet: /
File: Alfred.kicad_sch

Title: Alfred – one-servo thin client

Size: A4	Date: 2026-09-10	Rev: C – first article
KiCad E.D.A. 10.0.6		Id: 1/6

USB-C: fixed 5 V / 5 A contract

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Routed Alfred carrier. Unbuilt prototype; follow revision C bring-up checks.

Sheet: /USB-C: fixed 5 V / 5 A contract/
File: block1.kicad_sch

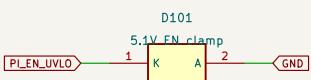
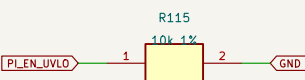
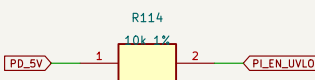
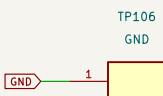
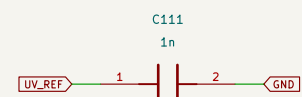
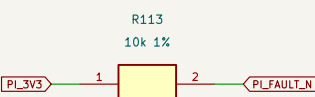
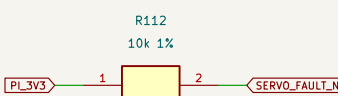
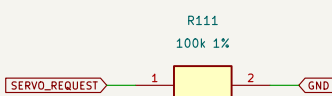
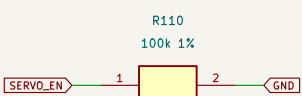
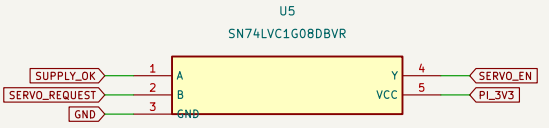
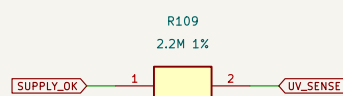
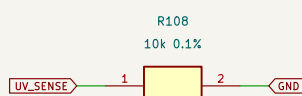
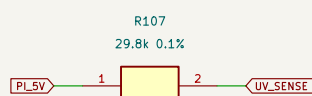
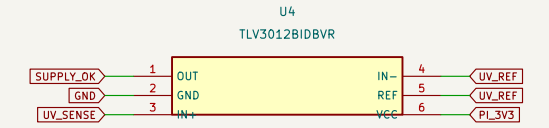
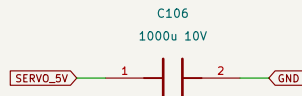
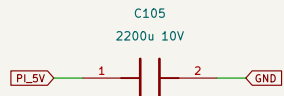
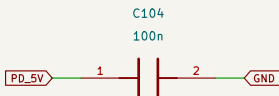
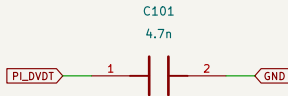
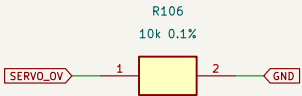
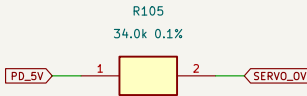
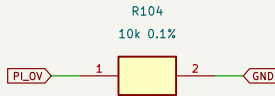
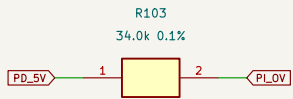
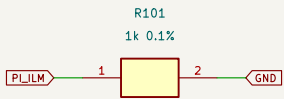
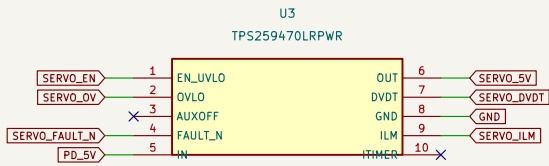
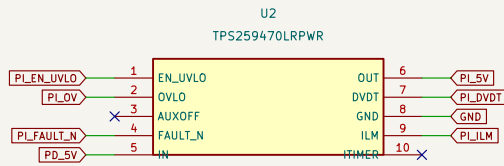
Title: USB-C: fixed 5 V / 5 A contract

Size: A2	Date: 2026-09-10
KiCad E.D.A. 10.0.6	

Rev: C – first article

Protected rails and servo undervoltage override

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Routed Alfred carrier. Unbuilt prototype; follow revision C bring-up checks.

A1-4RD / Alfred

Sheet: /Protected rails and servo undervoltage override/

File: block2.kicad_sch

Title: Protected rails and servo undervoltage override

Size: A2 Date: 2026-09-10

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Rev: C - first article

Id: 3/6

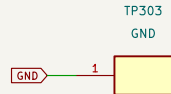
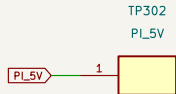
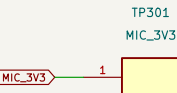
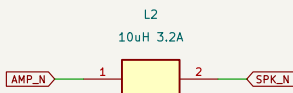
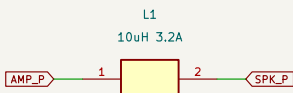
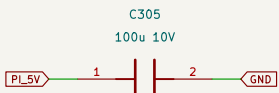
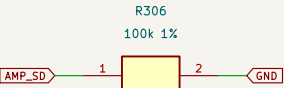
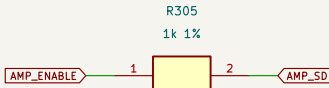
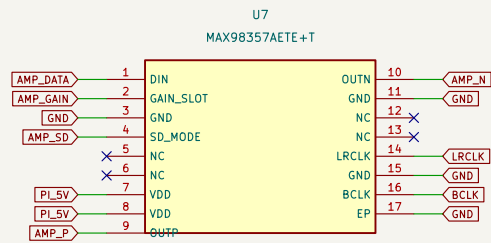
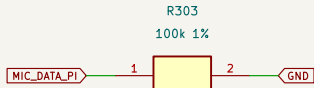
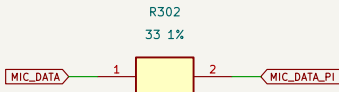
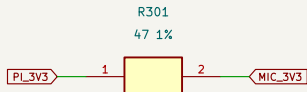
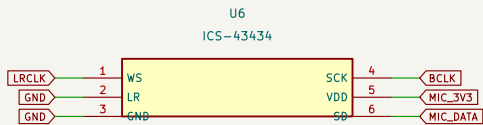
Raspberry Pi physical header and shared I2S

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Digital microphone and bridge speaker amplifier

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Routed Alfred carrier. Unbuilt prototype; follow revision C bring-up checks.

A1-4RD / Alfred

Sheet: /Digital microphone and bridge speaker amplifier/

File: block4.kicad_sch

Title: Digital microphone and bridge speaker amplifier

Size: A2

Date: 2026-09-10

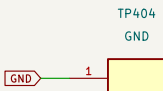
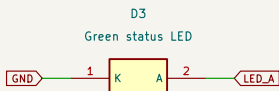
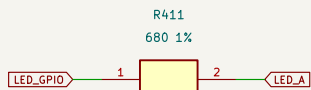
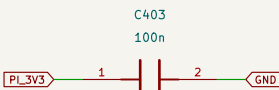
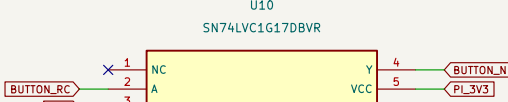
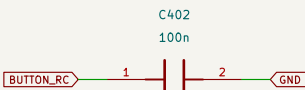
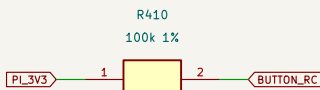
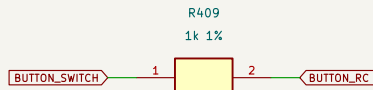
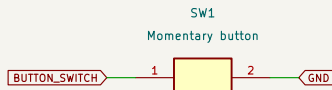
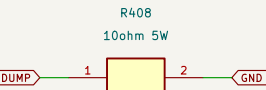
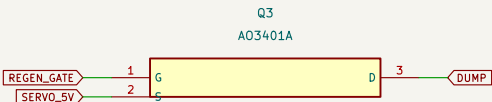
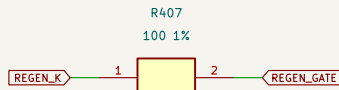
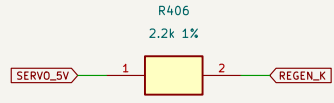
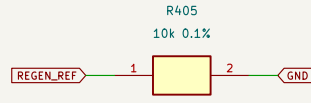
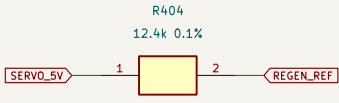
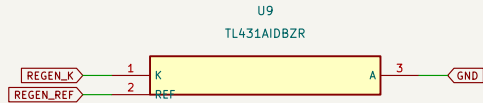
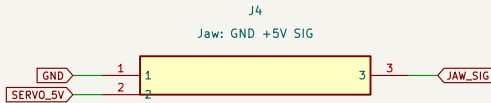
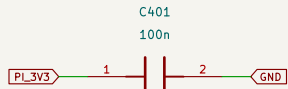
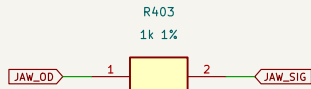
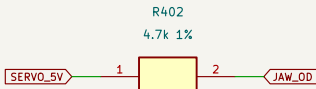
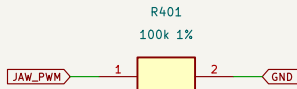
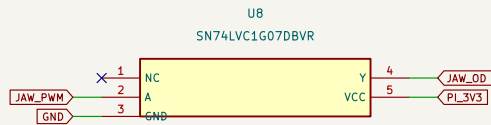
Rev: C - first article

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Jaw interface, energy dump, button and status

Net labels are global across sheets. Pin numbers match physical pads. NC pins explicitly marked.



Routed Alfred carrier. Unbuilt prototype; follow revision C bring-up checks.

A1-4RD / Alfred

Sheet: /Jaw interface, energy dump, button and status/

File: block5.kicad_sch

Title: Jaw interface, energy dump, button and status

Size: A2

Date: 2026-09-10

Rev: C - first article

KiCad E.D.A. 10.0.6

Id: 6/6